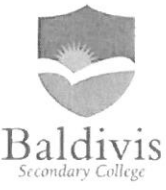


Name:			Date:	
Class:				
	Year 8 Mathematics 2019		/45	
	Test 4			
	Resources Allowed: calculator/ page of notes / ruler			
	Time allowed: 55 minutes			
<u>Full working out MUST be shown to get full marks for each question.</u>				

SECTION ONE: MULTIPLE CHOICE (circle the correct answer)

[5 marks]

1) What is the value of p in $p - 4 = -3$

- a. -7
- b. 7
- c. -1
- ☒ d. 1

2) What is another name for slope

- a. Intercept
- ☒ b. Gradient
- c. Axis
- d. Line

3) For the linear equation $y = -x + 5$. If x is 3 what is y

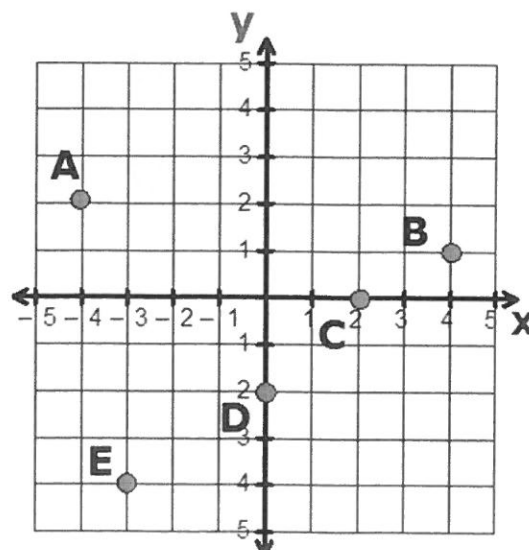
- a. 8
- b. -8
- ☒ c. 2
- d. -2

4) Using the graph, which point has a y value of 0

- a. A
- b. B
- c. C
- ☒ d. D

5) using the graph, which point has a x value of 0

- a. B
- ☒ b. C
- c. D
- d. E

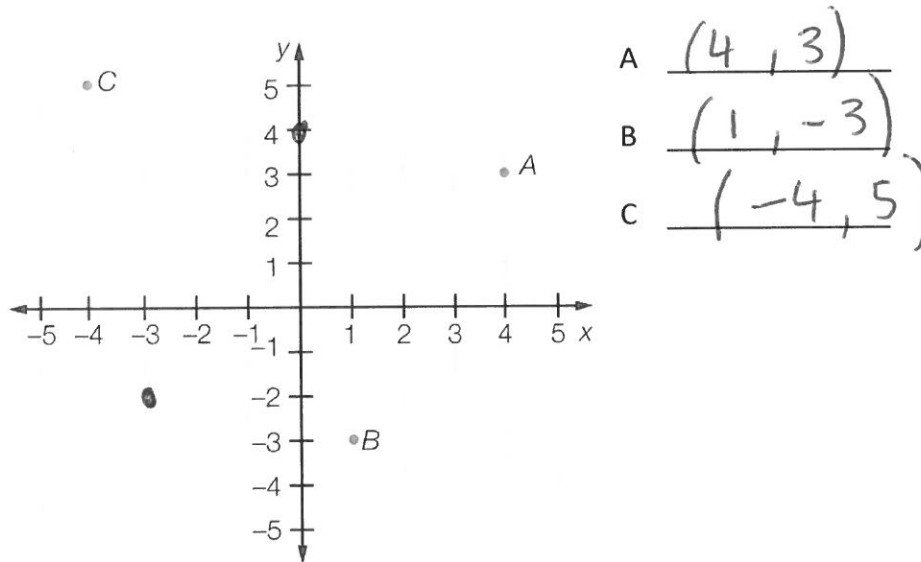


SECTION TWO: SHORT ANSWER QUESTIONS (show working out if necessary)

[40 marks]

- 6) Look at the Cartesian plane below and write down the coordinates of the points A, B and C

[5 marks]



Using the same Cartesian plane above place a ● on the coordinates $(-3, -2)$ and $(0, 4)$

- 7) **Circle** which of the following coordinates lie on the line $y = 10 - 3x$. Use mathematical reasoning to explain your choice. [2 marks]

- i. (1, 3)
 ii. (3, 1) ✓

Reason:

$$y = 10 - 3 \times 1$$

$$y = 7$$

$$y = 10 - 3 \times 3$$

$$y = 1$$

8) Use the rule $y = 4x + 2$ to fill in the table below then plot as a linear graph

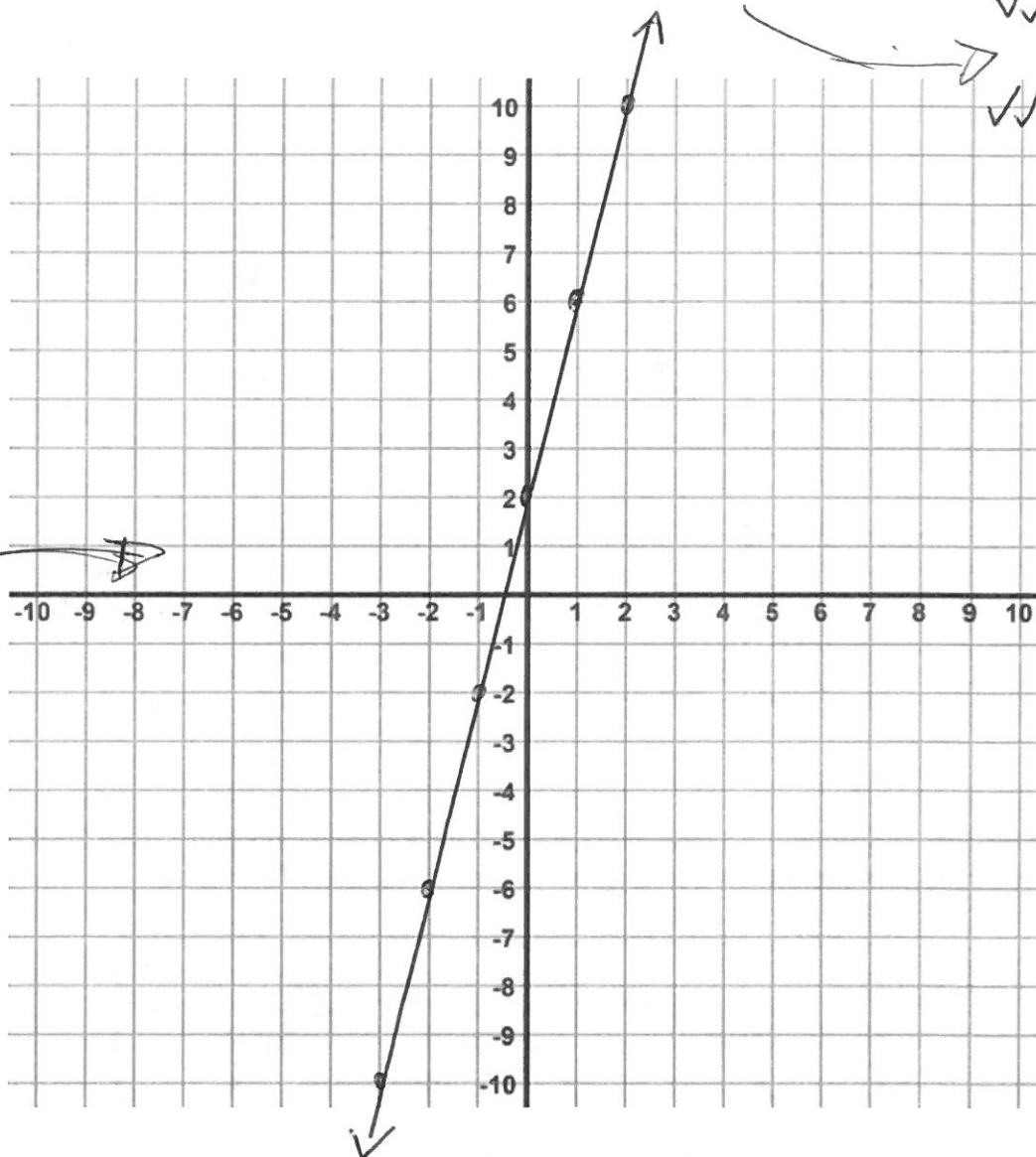
[6 marks]

x	-3	-2	-1	0	1	2	3
y	-10	-6	-2	2	6	10	14

✓ some correct

✓✓ most correct

✓✓✓ all correct



9) Find the rule for these tables of values.

[2 marks]

a

x	-2	-1	0	1	2
y	8	5	2	-1	-4

$$y = -3x + 2$$

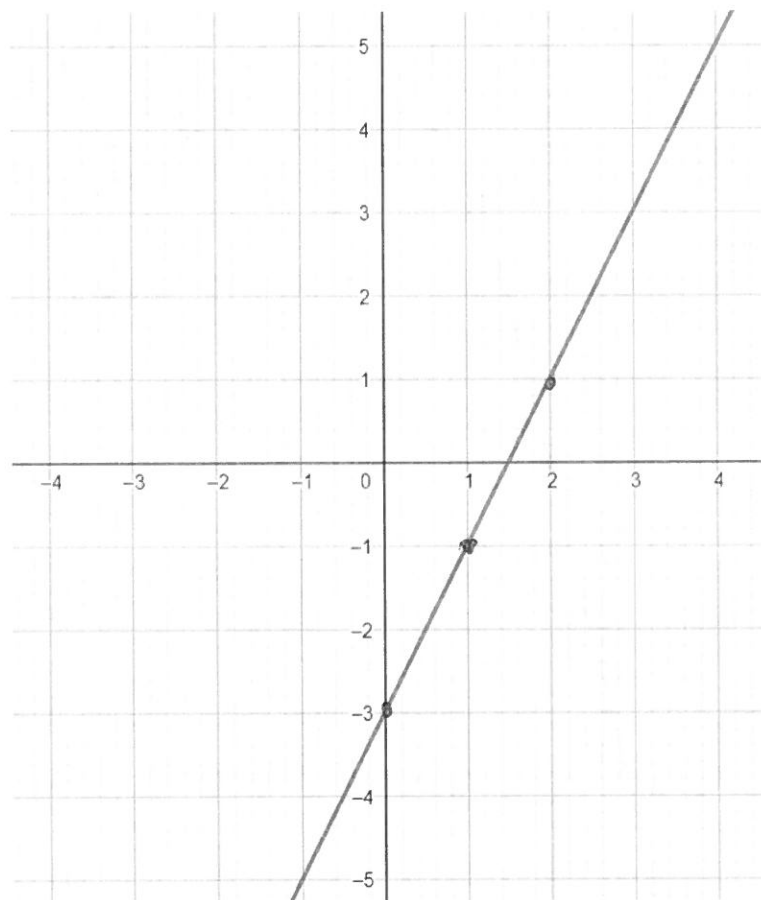
b

x	2	3	4	5	6
y	-1	1	3	5	7

$$y = 2x - 5$$

10) Find the equation of this line (complete a table of values to help you!)

[4 marks]



x	0	1	2	
y	-3	-1	1	

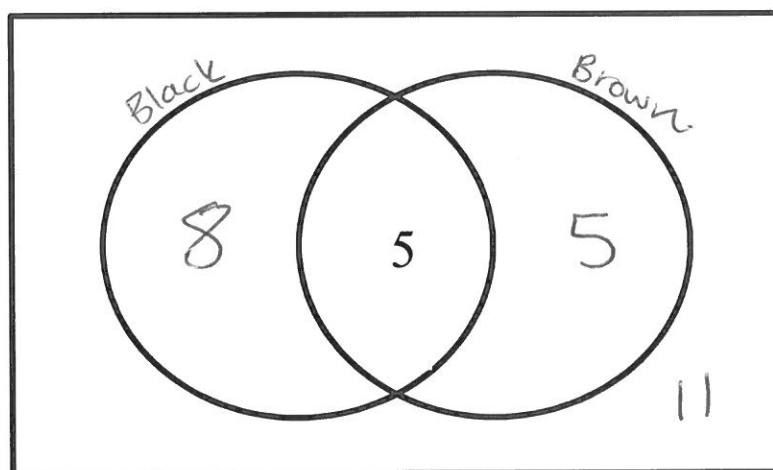
✓ - table used
✓✓ - table is correct!

$$y = 2x - 3$$

✓ ✓

11) In a class of 29 students, 13 have black hair, 10 have brown eyes and 5 have both black hair and brown eyes. Complete the Venn diagram representing this information.

[6 marks]



a. How many students do not have black hair? 16

b. How many students do not have brown eyes? 19

c. How many students have black hair or brown eyes? 18

- 12) A sample of teenagers between the ages of 16-19 were asked whether they passed or failed their driving test the first time. The results were further categorised by gender as shown [10 marks]

	Boys	Girls	Total
Passed	7	16	23
Failed	5	4	9
Total	12	20	32

- a. Complete the above two-way table
- b. How many teenagers participated in the survey? 32.
- c. How many of the teenagers surveyed failed their driving test? 9

- d. Find the **probability** that a randomly selected student:

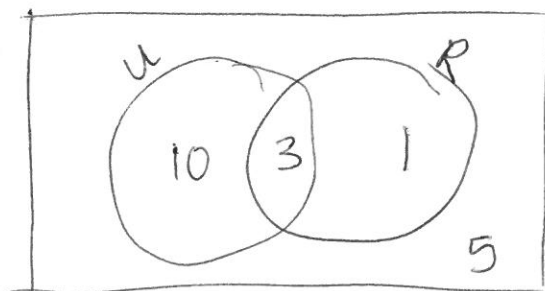
i. Is a girl who passed the test $\frac{16}{32}$

ii. Did not pass their driving test $\frac{9}{32}$

- e. What percentage of girls passed the test?

$$\frac{16}{32} \times 100 = 50\%$$

- 13) In a group of 19 people waiting at the bus stop one morning, 13 people had an umbrella, 4 people had a raincoat and 5 had neither an umbrella or a raincoat. Draw a Venn diagram or a two-way table to represent this information. [5 marks]



What is the **probability** that a randomly selected person:

i. Has a raincoat but not an umbrella? $\frac{1}{19}$

ii. Has an umbrella or a raincoat but not both? $\frac{11}{19}$

END OF TEST

